(20) $\sqrt{2}.(\sqrt{18}+\sqrt{50}-\sqrt{32+18}-\sqrt{36+100}-\sqrt{64}+\sqrt{16}$

$-6+10-8+4 = 12$

(21) $\frac{5\sqrt{16}}{\sqrt[5]{\frac{1}{2}}} \Rightarrow \sqrt[5]{\frac{16}{\frac{1}{2}}} \Rightarrow \sqrt[5]{32} = \sqrt[5]{2^5} = 2$

(22) $\sqrt{25-25x^2} - \sqrt{64-64x^2} \Rightarrow \sqrt{25(1-x^2)} - \sqrt{64(1-x^2)}$

$\Rightarrow 5\sqrt{1-x^2} - 8\sqrt{1-x^2}$

$\Rightarrow -3\sqrt{1-x^2}$

(23) $\sqrt[5]{\sqrt{\sqrt{\sqrt[3]{\sqrt{2}}}}} \Rightarrow \sqrt[5.2.2.3]{2} \Rightarrow \sqrt[120]{2}$

(24) $\sqrt[3]{2\sqrt[3]{4}} = \sqrt[3]{\sqrt[3]{2^3 \cdot 4}} = \sqrt[3.2]{2^4 \cdot 2^2} = \sqrt[6]{2^6} \cdot \sqrt[6]{2^4} = 2^{\frac{4}{3}}$

(25) $a>0$ olmak üzere $(\sqrt[5]{a^2})^4 \Rightarrow ?$

$(\sqrt[5]{a^2})^4 = \sqrt[5]{a^8} = \sqrt[5]{a^5 \cdot a^3} \Rightarrow a\sqrt[5]{a^3}$

(26) $\sqrt{(-4)^2} - \sqrt[3]{(-2)^3} + 4\sqrt{(-2)^4} + \sqrt[4]{1} \Rightarrow |-4| - (-2) + 4 \cdot 2 + 1 = 11$

(27) $a>0$ ve $b<0$ ise $\sqrt{(ba)^2} - \sqrt{(2a-b)^2} \Rightarrow ?$

$\sqrt{(b-a)^2} - \sqrt{(2a-b)^2} = |b-a| - |2a-b|$

$\Rightarrow -(b-a) - (2a-b)$

$\Rightarrow -b+a-2a+b$

$= -a$

(28) $\frac{2}{\sqrt{6}} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3}$

$\left(\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} + \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}\right) \cdot 2$

$\left((\sqrt{5}-\sqrt{3}) - (\sqrt{5}+\sqrt{3})\right) \cdot \frac{-(\sqrt{5}-\sqrt{3})}{\sqrt{5}+\sqrt{3}}$

(29) $A = \sqrt{5} - \sqrt{3}$ ve $B = \sqrt{5}+\sqrt{3}$ $\frac{A}{B} + \frac{B}{A} \cdot 2 = ?$

(10)

(13) $\left(\frac{a^x}{a^4}\right)^{x-4} \cdot \left(\frac{a^4}{a^x}\right)^{x-4}$ ifadesinin eşiti=?

$\hookrightarrow (a^{x-4})^{x-4} \cdot (a^{4-x})^{x-4}$

$\hookrightarrow (a^{x-4} \cdot a^{4-x})^{x-4} = (a^0)^{x-4} = 1^{x-4}$

(14) $3^x = 2$ göre $27^{x+1} - 81^x$ =?

$27^{x+1} - 81^x = (3^3)^{x+1} - (3^4)^x = 3^{x+3} - 3^{4x} \Rightarrow (3^x)^3 \cdot 3^3 - (3^x)^4$

$\Rightarrow 2^3 \cdot 3^3 - 2^4$

$\Rightarrow 200$

(15) $x-2=4$ olduğuna göre $|x-4| + |4-x|$ =?

$x-2=4 \quad x-4=-2 \quad 4-x=-2$

$2+1-21=4$

(16) $-2<x<4 \Rightarrow x+2 > 0 \quad x-4<0$

$x+2-x+4 = 6$

(17) $x<0<4$ ise $\frac{|x-4| - |4-x|}{x}$ =?

$x<0<4 \Rightarrow 2x<0 \quad 24>0$

$2x-4<0 \quad 4-x>0$

$\frac{-2x+4-24+x}{x} \Rightarrow \frac{-(x+4)}{x} = -1$

(18) $\sqrt{0,64} + \sqrt{0,04} - \sqrt[3]{0,027} \Rightarrow \sqrt{\frac{64}{100}} + \sqrt{\frac{4}{100}} - \sqrt[3]{\frac{27}{1000}}$

$\Rightarrow \sqrt{\left(\frac{8}{10}\right)^2} + \sqrt{\left(\frac{2}{10}\right)^2} - \sqrt[3]{\left(\frac{3}{10}\right)^3}$

$\Rightarrow \frac{8}{10} + \frac{2}{10} - \frac{3}{10} = \frac{7}{10} = 0,7$

(19) $\frac{\sqrt{75} - \sqrt{27} + \sqrt{12}}{\sqrt{12} + \sqrt{48}} \Rightarrow \frac{\sqrt{5^2 \cdot 3} - \sqrt{3^2 \cdot 3} + \sqrt{2^2 \cdot 3}}{\sqrt{6^2 \cdot 3} + \sqrt{4^2 \cdot 3}}$

$\Rightarrow \frac{4\sqrt{3}}{10\sqrt{3}} = \frac{2}{5}$

Üslü İfadeler

① $25^0 = 1$

② $\left(-\frac{2}{5}\right)^{-1} = -\frac{5}{2}$

③ $(-9)^0 = 1$

④ $-204^0 = -1$

⑤ $-\left[\left(-\frac{1}{2}\right)^{-1}\right]^4 = -\left(-\frac{1}{2}\right)^{-4} = -(-2)^4 = -16$

⑥ $-4^5 + 3.4^5 - 7.3^4.3 - 4^5 = (-1+3-1).3^4.(-7-3)$ =

⑦ $\frac{5^{10}+5^{15}+5^{20}}{5^{15}.5+5^{13}.5^{15}}$ $= x$ ise $\frac{x}{125}$ ifadesinin eşiti nedir?

⑧ $\frac{5^{10}}{5^{15}} = 5^{10-15} = 5^5$ $\frac{5^5}{5^3} = 5^2 = 25$

⑨ $\frac{16^n+6^n+2^n}{7^n+3^{n+1}} = \frac{(7.2)^n+(3.2)^n+2^n}{7^n+3^{n+1}+1} = \frac{7^n.2^n+3^n.2^n+2^n}{7^n+3^n.3-1} = 2^n$

⑩ $(0,027)^{\frac{5}{3}}.10^5$ işleminin sonucu nedir?

$(27.10^{-3})^{\frac{5}{3}}.10^5 = (3^3)^{\frac{5}{3}}.(10^{-3})^{\frac{5}{3}}.10^5 \Rightarrow 3^5.10^{-5}.10^5 = 3^5.10^0$

$\Rightarrow 3^5$

⑪ $\frac{-9^2.25^1}{125^{-1}.2^{-1}} \Rightarrow \frac{-(3^2)^{-2}.(5^2)^{-1}}{(5)^{3-1}.(3^3)^{-1}} = \frac{-3^{-4}.5^{-2}}{5^{-3}.3^{-3}} = -\frac{5^{-3+2}}{3^{-4+3}} = -\frac{5^{-1}}{3^{-1}}$

⑫ $\frac{x^{a+1}}{x^{a-2}} \cdot \left(\frac{x^{-a+2}.x^{2a+3}}{x}\right)^{-1} = ?$

0

$x^{a+1-a+2}.\left(x^{-a+2+2a+3-1}\right)^{-1} = x^3(x^{a+4})^{-1} = x^3.x^{-a-4} = x^{3-a-4}$